

Operating Systems

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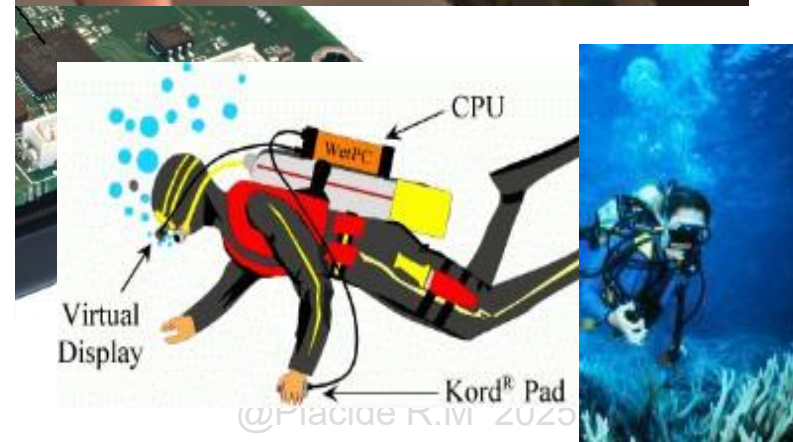
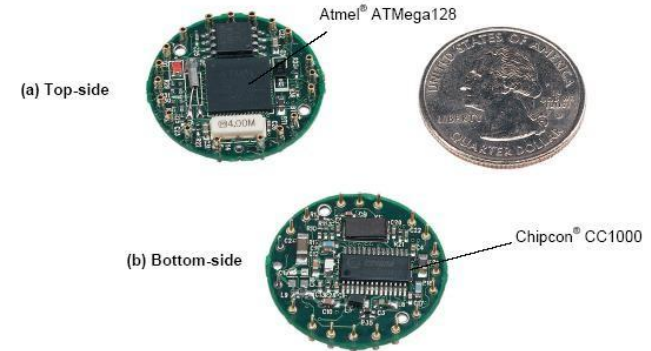
Course Description

- **Operating Systems** is a fundamental computer science course that introduces students to the principles, structures, and functions of operating systems. The course examines how operating systems **manage computer hardware and software** resources and provide an interface between users, application programs, and computer hardware.
- Students study important operating system concepts including **process management, CPU scheduling, threads, memory management, file systems, input/output management, storage management, synchronization, deadlocks, security, and virtualization**. The course combines theoretical concepts with practical exercises to help students understand how modern operating systems such as Linux and Windows manage system resources.

Why should I study Operating Systems?

- Need to understand interaction between the hardware and applications
 - New applications, new hardware..
- Need to understand basic principles in the design of computer systems
 - efficient resource management, security, flexibility
- Increasing need for specialized operating systems
 - e.g. embedded operating systems for devices
 - cell phones, sensors and controllers
 - real-time operating systems - aircraft control, multimedia services

©Lecture 1: Introduction to Operating Systems / Prof. Dr. J. K.



Course Objectives

The course aims to enable students to:

- Understand the role and functions of an operating system.
- Explain different operating system structures and architectures.
- Understand processes, threads, and process management.
- Apply CPU scheduling algorithms.
- Explain memory management and virtual memory.
- Understand file and storage management.
- Analyze synchronization and inter-process communication.
- Identify and handle deadlock situations.
- Explain operating system security and protection mechanisms.
- Gain practical experience with operating system administration and commands.

Main Course Topics

- 1. Introduction to Operating Systems**
- 2. Operating System Structures**
- 3. Process Management**
- 4. CPU Scheduling**
- 5. Process Synchronization**
- 6. Deadlocks**
- 7. Memory Management**
- 8. File-System Management**
- 9. Storage Management**
- 10. Input/Output Systems**
- 11. Operating System Security and Protection**
- 12. Virtualization and Modern Operating Systems**

1.Introduction to Operating Systems

- An Operating System (OS) is system software that manages a computer's hardware and software resources and provides services for application programs. It acts as an interface between the user, application software, and computer hardware.

Examples include:

- Microsoft Windows
- Linux
- macOS
- Android
- iOS

2. Main Functions of an Operating System

An operating system performs several important functions:

a. Process Management

Manages programs that are currently running, including creating, scheduling, and terminating processes.

b. Memory Management

Controls the allocation and use of the computer's main memory (RAM).

c. File Management

Creates, organizes, stores, retrieves, and protects files and directories.

d. Device Management

Controls hardware devices such as keyboards, printers, disks, displays, and network interfaces.

e. Security and Protection

Controls access to system resources and protects data from unauthorized access.

2. Main Functions of an Operating System

f. User Interface

Provides ways for users to interact with the computer through:

- Command-Line Interface (CLI)
- Graphical User Interface (GUI)

g. Resource Management

Allocates CPU time, memory, storage, and other resources among different programs and users.

3. Objectives of an Operating System

The major objectives are to:

1. Make the computer system convenient to use.
2. Manage hardware resources efficiently.
3. Provide a **reliable** environment for **application** programs.
4. Protect system resources and user data.
5. Provide mechanisms for communication between programs and hardware.

4. Evolution of Operating Systems

Generation	Main Characteristics
Early computers	Programs were executed manually
Batch systems	Jobs were collected and executed in batches
Multiprogramming	Several programs could reside in memory
Time-sharing	Multiple users could interact with the system
Personal computer OS	Designed for individual users
Network OS	Supports communication and resource sharing
Distributed OS	Coordinates resources across multiple computers
Mobile OS	Designed for smartphones and tablets
Modern/cloud systems	Support virtualization, containers, cloud computing, and distributed services

5. Types of Operating Systems

Batch Operating System – Executes groups of jobs with little or no user interaction.

Multiprogramming Operating System – Keeps several programs in memory so the CPU can switch between them.

Multitasking Operating System – Allows multiple applications to appear to run simultaneously.

Time-Sharing Operating System – Allows multiple users or processes to share CPU time interactively.

Real-Time Operating System (RTOS) – Provides responses within specified time constraints and is used in systems where timing is critical.

Distributed Operating System – Coordinates resources and activities across multiple interconnected computers.

Network Operating System – Provides services for computers connected through a network.

Mobile Operating System – Designed specifically for mobile devices.

6. Components of an Operating System

A typical operating system contains:

Kernel – The central part of the OS that directly manages hardware and system resources.

Process Manager – Manages processes and CPU scheduling.

Memory Manager – Manages RAM and virtual memory.

File-System Manager – Manages files and directories.

Device Manager – Controls hardware devices.

Security Manager – Provides protection and access control.

User Interface – Enables users to interact with the system.

7. Operating System as an Interface

The OS provides an interface between:

**User → Application Programs → Operating System
→ Hardware**

For example, when a user **saves** a document, the application requests the operating system to write the data to storage. The OS communicates with the appropriate hardware and manages the operation.

8. System Calls

A **system call** is a mechanism through which an application program requests a service from the operating system.

Common categories include:

- **Process control**
- **File management**
- **Device management**
- **Information maintenance**
- **Communication**
- **Protection**
- For example, a program may use a system call to open, read, write, or close a file.

9. Importance of Operating Systems

Operating systems are important because they:

- **Make computer systems easier to use.**
- **Manage hardware resources.**
- **Allow multiple applications to run.**
- **Provide security and protection.**
- **Manage files and storage.**
- **Facilitate communication between software and hardware.**
- **Improve system efficiency and reliability.**

key Summary

An operating system is the foundation of a computer system. It manages hardware resources, provides services to applications, controls program execution, manages files and memory, and provides security and user interfaces. Understanding operating systems is essential for studying areas such as computer architecture, networking, cybersecurity, cloud computing, and software development.

Assignment No 1

- **Discuss on computer system architecture**
- **Explain the Operating system structure**
- **Discuss on major function of operating system**

N.B Your work has to be :

- **Hand-written**
- **Well organized**
- **Deadline is next class**